

Daison Darlan

PhD Candidate in Artificial Intelligence

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📁 Code Repository

Research Profile

Modern AI has advanced rapidly, but still lacks the robustness and computational efficiency of biological intelligence. My work asks how this gap can be narrowed: what principles from biological learning can help artificial systems adapt, generalize, and remain stable under dynamic conditions? I approach this question by treating intelligence as an interaction between learning, memory, and regulation, not as task performance alone. Methodologically, I use reinforcement learning to model adaptive behavior, multi-objective optimization to study decision-making under competing goals, and neuro-inspired mechanisms to examine how artificial systems can regulate behavior and draw on prior experience. Together, these methods allow me to treat artificial systems not only as engineering tools for building more adaptive AI, but also as scientific models for probing the computational principles of biological intelligence.

This trajectory has shaped a research profile spanning optimization, NeuroAI, and applied intelligent systems, supported by first-author and collaborative publications across these areas. Beyond publications, I have contributed to government-funded research, technology transfer through patents and industry-facing projects, and academic service through peer review and mentoring.

Research Interests

NeuroAI | algorithms in nature | adaptive optimization | reinforcement learning | autonomous intelligent systems

Education

2022–2026 **Integrated MS–PhD in Artificial Intelligence**, *Kyungpook National University*, Daegu, South Korea.

Training focus: Developed a research foundation in artificial intelligence centered on optimization, reinforcement learning, and computer vision, while shaping a broader direction toward adaptive and neuro-inspired intelligent systems.

Status: Thesis successfully defended; degree conferral expected in August 2026.

GPA: 4.18/4.3

2017–2021 **B.Tech. in Aerospace Engineering**, *Amity University*, India.

Training focus: Trained in mathematical modeling, simulation, and engineering design, with emphasis on aerodynamics, computational analysis, and nature-inspired principles of efficient motion and stability.

GPA: 8.795/10

Research Experience

03/2022–08/2026 **PhD Research Fellow, Evolutionary Computation and Intelligent Systems Lab, Kyungpook National University**, *Daegu, South Korea*, Supervisor: Prof. Rammohan Mallipeddi.

Funded through the Kyungpook National University Kings Scholarship and National Research Foundation (NRF), Korea.

- Defined and developed a doctoral research direction on adaptive optimization for complex decision-making, focusing on how search processes can regulate behavior, reuse prior experience, and remain effective under competing objectives and constraints.
- Designed neuro-inspired algorithmic mechanisms that integrate regulation, memory reuse, and domain knowledge into evolutionary search, with applications to autonomous planning and

multi-objective decision-making.

- 03/2026– **Researcher, Ethical Stress Testing and Retrofit (E-STAR) for Autonomous Vehicles.**
08/2026 Funded by National Research Foundation (NRF), Korea.
- Addressed the challenge of testing autonomous vehicles under rare, uncertain, and safety-critical driving scenarios. Co-developed the funded NRF proposal and designed a framework using counterfactual scenario generation and evolutionary optimization to expose decision-making failures under real-world distribution shifts.
- 09/2025– **Researcher, Powerful Brain Project.**
03/2026 Funded by MSIT, Daegu Digital Innovation Promotion Agency (DIP), and National Information Society Agency (NIA), Korea.
- Explored how brain-derived feedback can inform computational models of reasoning and decision-making. Developed biologically guided reinforcement learning and post-training ideas aimed at improving adaptability, alignment, and efficiency in large language models.
- 04/2025– **Researcher, Domain-Adversarial Neural Network for HCP Face/Shape Classification.**
07/2025 Funded by the NRF Core Research Institute Basic Science Program, Ministry of Education, Korea (2021R1A6A1A03043144).
- Addressed cross-subject variability in fMRI-based face/shape classification, where individual neural differences can weaken model generalization. Designed a lightweight domain-adaptive model that aligned subject-specific feature distributions while preserving task-relevant neural information.
 - Achieved 96.62% test accuracy while outperforming heavier 3-D CNN and Transformer baselines with substantially lower computational cost.
- 06/2023– **Researcher, Development of Inclusive Unmanned Kiosk for Document Issuance Machine.**
06/2024 Funded through the Institute of Information & Communications Technology Planning & Evaluation (IITP), Korea government (MSIT), No. RS2023-00262841.
- Addressed accessibility barriers in unmanned public-service kiosks by developing lightweight AI modules for gaze, gesture, speech, and demographic analysis. These modules supported more inclusive interaction on low-resource computational systems.
 - Contributed to technology transfer, patent claims, KAIC certification, and journal publications arising from the project.
- 09/2023– **Researcher, Development of a Circulating, Single-drop Nutrient Solution and Cloud Control Platform Using AI-based Strawberry Growth Stage Recognition.**
12/2023 Funded by the Digital Innovation Hub (DIP, MSIT & Daegu City, 2023, DBSD2-01) and the National Information Society Agency (NIA, MSIT, 2022-3-030).
- Addressed the need for automated crop monitoring in smart greenhouses by building a strawberry growth-stage dataset and benchmarking deep learning models across seven developmental stages. The system supported stage-aware precision nutrient management through visual growth recognition.
- 01/2023– **Researcher, A Real-time Warning System Using Deep Learning-based Vehicle Identification and Tracking Technology to Prevent Traffic Accidents.**
10/2023 Funded through Daegu Techno Park (DGTP) 2023 Regional Innovation Centered University Research Activity Support Project.
- Addressed real-time vehicle identification and tracking for CCTV-based accident-prevention systems. Developed semi-supervised vehicle tracking and re-identification models using synthetic data to reduce annotation dependence, supporting technology transfer and patent publication.

Selected Research Outputs (*Google Scholar*)

My research outputs reflect three connected strands: adaptive optimization for complex decision-making, biologically inspired approaches to artificial intelligence, and applied systems where robustness and generalization matter. The selected publications below emphasize first-author contributions and collaborative work that support this trajectory.

A. *Journal Articles*

- 2026 **Daison Darlan**, Oladayo S Ajani, and Rammohan Mallipeddi. Evolutionary algorithm with domain-specific operators for UAV path planning. *Swarm and Evolutionary Computation*, volume 101, page 102267. Elsevier, 2026, **(SCIE : Q1)**.
- 2026 Ivan Fenyom, Adeyinka Adedigba, **Daison Darlan**, Oladayo S Ajani, Rammohan Mallipeddi, and Hwang Jae Joon. Lightweight webcam-based eye tracking system for large display screens. *Multimedia Tools and Applications*, volume 85, page 12. Springer, 2026, **(SCIE : Q2)**.
- 2026 Oladayo S Ajani, **Daison Darlan**, Esther Aboyeji, Kalyana C Veluvolu, and Rammohan Mallipeddi. Dynamic hand exercise recognition for game-based finger rehabilitation. *Scientific Reports*. Nature Publishing Group UK London, 2026, **(SCIE : Q1)**.
- 2025 **Daison Darlan**, Oladayo S Ajani, Anand Paul, and Rammohan Mallipeddi. A multi-objective benchmark for UAV path planning with baseline results. *Swarm and Evolutionary Computation*, volume 96, page 101968. Elsevier, 2025, **(SCIE : Q1)**.
- 2025 **Daison Darlan**, Oladayo S Ajani, Joon Woo An, Nan Yeon Bae, Bram Lee, Tusan Park, and Rammohan Mallipeddi. SmartBerry for AI-based growth stage classification and precision nutrition management in strawberry cultivation. *Scientific Reports*, volume 15, page 14019. Nature Publishing Group UK London, 2025, **(SCIE : Q1)**.
- 2025 Oladayo S Ajani, Ivan Fenyom, **Daison Darlan**, and Rammohan Mallipeddi. Prediction-guided multi-objective reinforcement learning with corner solution search. *Computers and Electrical Engineering*, volume 122, page 109964. Elsevier, 2025, **(SCIE : Q1)**.
- 2024 Oladayo S Ajani, Dzeuban Fenyom Ivan, **Daison Darlan**, Ponnuthurai N Suganthan, Kaizhou Gao, and Rammohan Mallipeddi. Deep reinforcement learning as multiobjective optimization benchmarks: Problem formulation and performance assessment. *Swarm and Evolutionary Computation*, volume 90, page 101692. Elsevier, 2024, **(SCIE : Q1)**.

B. *International Conference Proceedings*

- 2026 Shreyan Kundu, Souradeep Mukhopadhyay, **Daison Darlan**, and Rammohan Mallipeddi. Is negative correlation learning a solution for cross domain colon polyp segmentation? In *Eighteenth International Conference on Machine Vision (ICMV 2025)*, volume 14114, pages 118–125. SPIE, 2026.
- 2025 Shreyan Kundu, Souradeep Mukhopadhyay, **Daison Darlan**, and Rammohan Mallipeddi. AURA-Net: Adaptive Uncertainty-Weighted Ranking and Attention-Driven Network for Generalized Colon Polyp Segmentation. In *Pacific Rim International Conference on Artificial Intelligence*, pages 20–35. Springer, 2025.
- 2024 Vaishnavi Vats, Ivan Fenyom, Oladayo S Ajani, **Daison Darlan**, and Rammohan Mallipeddi. EEG-Based Emotion Classification Through Multi-Objective Hyperparameter Search. In *International Conference on Computing, Intelligence and Data Analytics*, pages 383–395. Springer, 2024.
- 2023 **Daison Darlan**, Oladayo S Ajani, Victor Parque, and Rammohan Mallipeddi. Recognizing social touch gestures using optimized class-weighted CNN-LSTM networks. In *2023 32nd IEEE international conference on robot and human interactive communication (RO-MAN)*, pages 2024–2029. IEEE, 2023.
- 2023 **Daison Darlan**, Oladayo S Ajani, and Rammohan Mallipeddi. Lunar landing site selection using machine learning. In *2023 International Conference on Machine Intelligence for GeoAnalytics and Remote Sensing (MIGARS)*, volume 1, pages 1–4. IEEE, 2023.

- 2026 **Daison Darlan**, Oladayo S Ajani, Adeyinka Adedigba, and Rammohan Mallipeddi. Experience replay and operator plasticity in evolutionary uav path planning. *Expert Systems with Applications; Under Review*, 2026, (**SCIE : Q1**).
- 2026 Shreyan Kundu, Sauradeep Mukhopadhyay, Semanti Das, Basu Mallik, **Daison Darlan**, and Rammohan Mallipeddi. A3FUS-Net: Towards Automated Renal Cell Carcinoma Grading System via Advanced Adaptive Attention Mechanism aided Deep Learning Fusion. *Computers and Electrical Engineering; In Press*, 2026, (**SCIE : Q1**).

Patents

- 2023 **No.C-2023-051898**, Korea Alpha System Co., Ltd, Rammohan Mallipeddi, Oladayo S. Ajani, **Daison Darlan**, and Dzeuban Fenyom Ivan , Anomalous Vehicle Detection and Alerting System using CCTV Camera Network. November, 2023.

Fellowships and Awards

- 2025 **G-KNU Academic Scholarship**, 100% tuition waiver, Kyungpook National University.
- 2022–2024 **KINGS Academic Scholarship**, 100% tuition waiver, Kyungpook National University.
- 2022–2023 **Brain Korea 21 Fellowship**, National Research Foundation of Korea.
- 2018–2019 **Academic Scholarship**, 50% tuition waiver, Amity University.
- 2017–2018 **Academic Scholarship**, 100% tuition waiver, Amity University.

Teaching and Mentoring

Mentored junior collaborators in research coding, experimental design, and manuscript preparation. Supported undergraduate and graduate students as a Graduate Teaching Assistant in core artificial intelligence subjects.

- Fall 2025 Deep Learning
- Spring 2024 Reinforcement Learning 101
- Fall 2023 Mathematics for AI 102
- Spring 2023 Mathematics for AI 101

Technical and Methodological Expertise

- Adaptive AI and Optimization Designing optimization-driven learning systems for constrained, multi-objective, and uncertain environments, including evolutionary algorithms, reinforcement learning, and adaptive search mechanisms.
- Bio-inspired Learning Translating biological principles such as regulation, memory, and representation into computational frameworks for adaptive intelligence and neural data analysis.
- Applied AI Building and evaluating deep learning systems for autonomous planning, biomedical AI, visual recognition, tracking, pose analysis, agriculture, and human-centered technologies.
- Research Software Developing reproducible research code in Python, MATLAB, and C++; designing experiments, automating training pipelines, and producing publication-quality analysis .

Academic Service

- Peer Review (ORCID) Contribute to academic peer review across artificial intelligence, evolutionary computation, optimization, and engineering systems, including reviews for *Engineering Applications of Artificial Intelligence*, *Swarm and Evolutionary Computation*, *Information Sciences*, *Computers and Electrical Engineering*, and *IEEE Transactions on Systems, Man, and Cybernetics* among others.